

Through the Lower Critical Endpoint

Exponentially Coalescing Folds,
Landau Transseries,
and Conditional Action Budgets

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29 September 2026

Abstract

We study the lower-endpoint problem for the countable-action equation $U = c \operatorname{Li}_{2+\varepsilon}(qe^U)$, where $\alpha = 1 + \varepsilon \downarrow 1$. On the interior-fold side, an exact saddle normalization removes the apparent pole at $\varepsilon = 0$. It yields a uniform coefficient theorem with no restriction on $\varepsilon \log(1/\delta)$, where $e^{-\delta}$ is the fold location. At the endpoint, $\delta \sim e^{-1/c}$, and the crossover variable is $\lambda = nc\delta$. The limiting coefficient profile is an explicit transform of the classical Landau density. We prove its all-orders finite-coefficient correction scheme, a convergent Lambert inverse chart and a uniform two-sheet fold chart. We also identify the first obstruction to extending the compact-crossover formula into the Gaussian range: a relative factor $\exp(\lambda\delta/4)$.

For a truncation to actions $j \leq M$, we obtain the exact limiting conditional profile and prove that coefficient preservation is equivalent to $M\delta \rightarrow \infty$ in a compact crossover window. Its loss has a sharp doubly exponential equivalent, including the prefactor. Consequently the scaled action budget for a small limiting loss h grows like $\lambda \log \log(1/h)$, not like $\log(1/h)$. Proofs, exact algebraic checks, numerical diagnostics, a source audit, and further research questions are included.

Scope. This is a conventional mathematical research draft. It resolves a specified interior-fold part of a repository-local lower-endpoint question, not every critical-coupling path at $\alpha = 1$. The classical Landau law, Lagrange inversion, and Poisson insertion identities are not claimed as new. Global priority, independent refereeing, and Lean verification are not asserted.

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1 The question and the contribution

1.1 The repository boundary being addressed

The `ProveIt` package *Beyond Finite-Action Folds* studies positive countable-action equations with tail $j^{-1-\alpha}$ for $1 < \alpha < 2$. Its later editorial cross-references record work at the upper endpoint and explicitly leave the lower endpoint $\alpha \rightarrow 1$ unresolved [1, 2]. The present paper addresses the pure-tail family

$$U_{\varepsilon,c}(q) = c \sum_{j \geq 1} \frac{q^j e^{j U_{\varepsilon,c}(q)}}{j^{2+\varepsilon}} = c \text{Li}_{2+\varepsilon}(q e^{U_{\varepsilon,c}(q)}), \quad c > 0, \quad \varepsilon \geq 0. \quad (1.1)$$

Its tail exponent is $\alpha = 1 + \varepsilon$. The ordinary coefficients are $u_n(\varepsilon, c) = [q^n]U_{\varepsilon,c}(q)$; they are not multiplied by $n!$.

For $\varepsilon > 0$, the boundary-critical coupling is $c_b = 1/\zeta(1 + \varepsilon)$. For $c > c_b$, the physical branch encounters an interior fold before the polylogarithmic boundary. At $\varepsilon = 0$, every $c > 0$ has such a fold. As $c \downarrow 0$, its location approaches the boundary at an exponentially small distance. This singular approach is not captured by putting $\varepsilon = 0$ separately in terms containing $\Gamma(-1 - \varepsilon)$ and $\zeta(1 + \varepsilon)$.

We normalize by the *exact* fold location before taking a limit. This produces one formula for all relative rates at which ε and the fold distance vanish. It also exposes a conditional large-jump phenomenon: a very large positive jump must be compensated by an exceptionally unlikely negative fluctuation of the remaining centered cloud. That compensation changes the action-cutoff loss from an ordinary exponential tail into a doubly exponential one.

Editorial note (ProveIt, 2026-09-29). The same lower endpoint is posed, independently, as Question 10 of the confluent package (`Confluent_Critical_Transseries_Exponent_Two_Boundary/Confluent_Critical_Transseries.tex`, “The lower endpoint $\alpha \downarrow 1$ ”: a compensated core, and whether conditioning couples the largest actions to the total) and as Question 8 of the logarithmic-endpoint package (`Logarithmic_Critical_Endpoint_Lambert_Charts/article.tex`, “The lower endpoint and loss of the mean”: a renormalized coupling, critical coordinate and action budget). Both were already in the repository at the commit this article was written against. For the pure tail on the interior-fold side, Theorem 3.1 supplies the compensated core and the coordinates δ and $\lambda = nA\delta$; Corollary 8.2 supplies the budget; and Theorem 9.2 shows that conditioning couples a large action to a doubly exponentially unlikely compensating fluctuation. The boundary-critical side of both questions is not addressed here (Section 11). Paths are relative to `Analysis/Transseries/docs/series-and-transseries/`.

1.2 What is inherited and what is established here

The substitution $t = qe^U$, its Lagrange coefficient identity, and the conditional-Poisson representation already occur in the repository’s critical-Hahn article [1]. More generally, the connection with simply generated trees and conditioned allocations is classical; Janson gives an extensive treatment [3]. The polylogarithm continuation formula is classical [4, Eq. 25.12.12]. The Landau distribution is likewise an established stable law, not a new special function introduced by this paper [5].

The results proved below are the following linked, model-specific statements. The distinction between a proved statement in this draft and an established claim of publication priority is maintained throughout.

Result	Added content and boundary
Exact confluence	A convergent normal form uniform through $\varepsilon = 0$, with no bound on $\varepsilon \log(1/\delta)$.
Coefficient crossover	A uniform additive error $O(\delta^{1-\eta})$ for $0 \leq \varepsilon \leq \eta < 1/2$ and compact positive λ -windows; an all-orders expansion at $\varepsilon = 0$.
Inverse charts	A genuinely convergent exponential-scale inverse and a uniform two-sheet chart through the moving fold.
Matching obstruction	The factor $e^{\ell/4}$ when $\lambda \rightarrow \infty$ and $\lambda\delta \rightarrow \ell$; the compact-window theorem is not silently used outside its range.
Conditional budget	A limiting cutoff profile, an if-and-only-if preservation criterion, and a sharp doubly exponential tail with its prefactor.

1.3 Limits that are deliberately not identified

Three different limiting statements will appear. A *coefficient crossover* lets $n \rightarrow \infty$, $\delta \rightarrow 0$, and possibly $\varepsilon \rightarrow 0$, with $\lambda = nA\delta$ in a compact subset of $(0, \infty)$. A *profile asymptotic* subsequently lets λ tend to zero or infinity, or the scaled cutoff m tend to infinity. A *Gaussian matching theorem* has its own proof for $\lambda \rightarrow \infty$. Only the last of these justifies a simultaneous unbounded- λ limit. Similarly, the small-tolerance action budget at the end of the paper is a statement about the limiting profile, not an unproved finite- n certificate at an arbitrarily small tolerance.

2 Exact coordinates and coefficients

Fix $0 \leq \varepsilon \leq \eta < 1/2$. For $\delta > 0$, define

$$\tau = e^{-\delta}, \quad c = \frac{1}{\text{Li}_{1+\varepsilon}(e^{-\delta})}, \quad (2.1)$$

$$A = c\Gamma(1 - \varepsilon)\delta^\varepsilon, \quad \lambda = nA\delta. \quad (2.2)$$

All these parameters are positive. The choice (2.1) parametrizes the entire interior-fold side for $\varepsilon > 0$, and all positive couplings for $\varepsilon = 0$. Introduce

$$Q(x) = -x - c\text{Li}_{2+\varepsilon}(e^{-x}), \quad (2.3)$$

$$\rho_0 = \exp\{-c\zeta(2 + \varepsilon)\}, \quad \rho = \exp\{Q(\delta)\}. \quad (2.4)$$

The notation $Q(0)$ denotes the continuous boundary value $\log \rho_0$. Unless otherwise stated, complex logarithms are principal.

Proposition 2.1 (Exact real fold). *Under (2.1), $Q'(\delta) = 0$ and*

$$Q''(\delta) = -c\text{Li}_\varepsilon(e^{-\delta}) < 0.$$

The physical real branch is parametrized by

$$t = qe^U, \quad U = c\text{Li}_{2+\varepsilon}(t), \quad q = t \exp\{-c\text{Li}_{2+\varepsilon}(t)\}, \quad 0 \leq t \leq \tau.$$

At $\varepsilon = 0$, one has the exact identities

$$\tau = 1 - e^{-1/c}, \quad \delta = -\log(1 - e^{-1/c}), \quad A = c. \quad (2.5)$$

Consequently $\delta \sim e^{-1/c}$ as $c \downarrow 0$.

Proof. Differentiating the polylogarithm series inside its disk of convergence and then substituting $t = e^{-x}$ gives $Q'(x) = -1 + c \operatorname{Li}_{1+\varepsilon}(e^{-x})$ and $Q''(x) = -c \operatorname{Li}_{\varepsilon}(e^{-x})$. The latter is negative on the positive real axis. Equation (2.1) therefore gives the unique stationary point. Equivalently, $t \mapsto te^{-c \operatorname{Li}_{2+\varepsilon}(t)}$ is strictly increasing up to $t = \tau$. The formal solution at zero is unique by coefficient induction and is analytic there by the implicit function theorem. The displayed parametrization continues its positive real branch to the fold. Finally, $\operatorname{Li}_1(t) = -\log(1-t)$ turns the fold equation into $-c \log(1-\tau) = 1$, proving (2.5). $\square$

Proposition 2.2 (Classical coefficient and conditioning identities). *For every integer $n \geq 1$,*

$$u_n(\varepsilon, c) = \frac{1}{n} [t^n] \exp\{nc \operatorname{Li}_{2+\varepsilon}(t)\}. \quad (2.6)$$

Let N_j be independent Poisson variables with means

$$\mu_j = nc \frac{e^{-\delta j}}{j^{2+\varepsilon}}, \quad S = \sum_{j \geq 1} j N_j. \quad (2.7)$$

Then $\mathbb{E}S = n$, and

$$u_n = \frac{\rho^{-n}}{n} \mathbb{P}(S = n). \quad (2.8)$$

If $u_{n,M}$ is the coefficient for the equation with actions $1 \leq j \leq M$, then

$$\frac{u_{n,M}}{u_n} = \mathbb{P}(N_j = 0 \text{ for every } j > M \mid S = n). \quad (2.9)$$

In particular this ratio is in $[0, 1]$, increases with M , and is exactly one for $M \geq n$.

Proof. Put $t = q \exp(c \operatorname{Li}_{2+\varepsilon}(t))$. Lagrange inversion gives

$$[q^n] c \operatorname{Li}_{2+\varepsilon}(t(q)) = \frac{1}{n} [t^{n-1}] c \operatorname{Li}'_{2+\varepsilon}(t) e^{nc \operatorname{Li}_{2+\varepsilon}(t)} = \frac{1}{n} [t^n] e^{nc \operatorname{Li}_{2+\varepsilon}(t)}.$$

The last equality follows by differentiating the exponential and comparing its coefficient of degree $n-1$. The generating function of S is

$$\mathbb{E}z^S = \exp\{nc(\operatorname{Li}_{2+\varepsilon}(\tau z) - \operatorname{Li}_{2+\varepsilon}(\tau))\}.$$

Its coefficient of z^n , together with $\tau = e^{-\delta}$, yields (2.8). The mean is $nc \operatorname{Li}_{1+\varepsilon}(\tau) = n$. The total Poisson intensity is finite, so the random sum is well defined. Requiring no jump above M replaces the positive-degree part of the generating exponent by its truncation while retaining the *full* zero-event factor. Dividing the resulting coefficient by the unrestricted one proves (2.9). These identities are included for clarity, not claimed as new results. $\square$

3 A convergent normal form through the lower endpoint

Define, for $\operatorname{Re} s > 0$,

$$f_{\varepsilon}(s) = \begin{cases} \frac{s - s^{1+\varepsilon}/(1+\varepsilon)}{\varepsilon}, & \varepsilon \neq 0, \\ s(1 - \log s), & \varepsilon = 0. \end{cases} \quad (3.1)$$

The singularity at $\varepsilon = 0$ is removable. Direct differentiation gives

$$f'_{\varepsilon}(s) = \frac{1 - s^{\varepsilon}}{\varepsilon}, \quad f'_{\varepsilon}(1) = 0, \quad f''_{\varepsilon}(1) = -1, \quad f_{\varepsilon}(1) = \frac{1}{1+\varepsilon}, \quad (3.2)$$

with the continuous interpretation of the first expression.

Theorem 3.1 (Exact confluent normal form). *For $0 \leq \varepsilon \leq \eta < 1/2$, $0 < \delta < 2\pi$, and $\operatorname{Re} s > 0$, $|\delta s| < 2\pi$,*

$$F_{\varepsilon,\delta}(s) := \frac{Q(\delta s) - Q(0)}{A\delta} = f_{\varepsilon}(s) + R_{\varepsilon,\delta}(s), \quad (3.3)$$

where the following series is convergent:

$$R_{\varepsilon,\delta}(s) = \frac{1}{\Gamma(1-\varepsilon)} \sum_{k \geq 2} \frac{(-1)^k \zeta(2+\varepsilon-k)}{k!} \delta^{k-1-\varepsilon} (ks - s^k). \quad (3.4)$$

For every compact $K \subset \{\operatorname{Re} s > 0\}$,

$$\sup_{s \in K} |R_{\varepsilon,\delta}(s)| \leq C_{K,\eta} \delta^{1-\varepsilon} \leq C_{K,\eta} \delta^{1-\eta} \quad (0 < \delta \leq \delta_K). \quad (3.5)$$

The same bound holds for each fixed number of derivatives in s . In particular no assumption on $\varepsilon \log(1/\delta)$ is made.

Proof. For nonintegral order, the classical convergent continuation formula is

$$\operatorname{Li}_{2+\varepsilon}(e^{-x}) = \Gamma(-1-\varepsilon)x^{1+\varepsilon} + \sum_{k \geq 0} \zeta(2+\varepsilon-k) \frac{(-x)^k}{k!}, \quad |x| < 2\pi, \quad |\arg x| < \pi. \quad (3.6)$$

This is [4, Eq. 25.12.12]; the separate singular terms at integral order must first be combined. The fold equation, using its derivative version, says

$$c\zeta(1+\varepsilon) - 1 = -c\Gamma(-\varepsilon)\delta^\varepsilon - c \sum_{j \geq 1} \zeta(1+\varepsilon-j) \frac{(-\delta)^j}{j!}.$$

Insert this equality into the linear term of $Q(\delta s) - Q(0)$. The gamma identities

$$-\Gamma(-\varepsilon) = \frac{\Gamma(1-\varepsilon)}{\varepsilon}, \quad \Gamma(-1-\varepsilon) = \frac{\Gamma(1-\varepsilon)}{\varepsilon(1+\varepsilon)}$$

produce f_{ε} . Reindexing the remaining linear sum with $k = j + 1$ produces exactly $ks - s^k$, with the sign shown in (3.4).

After this combination, every coefficient in (3.4) is regular for $0 \leq \varepsilon \leq \eta$; the zeta arguments there stay away from its pole. The power series is normally convergent when δ and δs stay in smaller disks. Passing to $\varepsilon = 0$ is therefore legitimate. For completeness, uniform geometric coefficient bounds can be obtained from the zeta functional equation and Stirling's formula: for any $r < 2\pi$, the quantities $|\zeta(2+\varepsilon-k)|/k!$ are bounded by $C_{r,\eta} r^{-k}$ uniformly in ε . Alternatively they follow by Cauchy estimates for the regular analytic remainder of (3.6) on a smaller disk. Factoring out $\delta^{1-\varepsilon}$ now proves (3.5); differentiated series give the derivative bounds. $\square$

Corollary 3.2 (Endpoint coefficients and fold value). *At $\varepsilon = 0$, $R_{0,\delta}$ is analytic in δ , and*

$$R_{0,\delta}(s) = \delta \left(-\frac{s}{2} + \frac{s^2}{4} \right) + \frac{\delta^2}{72} (3s - s^3) - \frac{\delta^4}{14400} (5s - s^5) + O_K(\delta^6), \quad (3.7)$$

$$F_{0,\delta}(1) = 1 - \frac{\delta}{4} + \frac{\delta^2}{36} - \frac{\delta^4}{3600} + O(\delta^6), \quad (3.8)$$

$$F_{0,\delta}''(1) = -\frac{\delta}{e^\delta - 1}. \quad (3.9)$$

Here $F_{0,\delta}'(1) = 0$ holds exactly, not just asymptotically.

Proof. Use $\zeta(0) = -1/2$, $\zeta(-1) = -1/12$, $\zeta(-2r) = 0$ for positive integers r , and $\zeta(-3) = 1/120$ in (3.4). Every summand has derivative zero at $s = 1$. Finally, $F''_{0,\delta}(1) = \delta Q''(\delta)/c = -\delta \text{Li}_0(e^{-\delta})$, proving (3.9). $\square$

Proposition 3.3 (All rates of confluence). *As $\varepsilon \downarrow 0$ and $\delta \downarrow 0$, jointly,*

$$\frac{1}{c} = \frac{1 - \delta^\varepsilon}{\varepsilon} + O(1), \quad c \sim \frac{\varepsilon}{1 - \delta^\varepsilon}, \quad (3.10)$$

with the quotients interpreted continuously at $\varepsilon = 0$. If $L = \log(1/\delta)$ and $r = \varepsilon L$, then $c \sim 1/L$ for $r \rightarrow 0$, $c \sim \varepsilon/(1 - e^{-r_0})$ for $r \rightarrow r_0 \in (0, \infty)$, and $c \sim \varepsilon$ for $r \rightarrow \infty$.

Proof. The derivative form of (3.6) gives

$$\text{Li}_{1+\varepsilon}(e^{-\delta}) = \zeta(1 + \varepsilon) - \frac{\Gamma(1 - \varepsilon)}{\varepsilon} \delta^\varepsilon + O(\delta).$$

The functions $\zeta(1 + \varepsilon) - 1/\varepsilon$ and $(\Gamma(1 - \varepsilon) - 1)/\varepsilon$ are bounded near zero. This proves the first formula, uniformly since $0 \leq \delta^\varepsilon \leq 1$. Its leading term is $L(1 - e^{-r})/r$, which tends to infinity for every joint approach: when $r \leq 1$ it is comparable to L , and when $r \geq 1$ it is comparable to $1/\varepsilon$. The additive $O(1)$ is therefore relatively negligible. The stated regimes follow. $\square$

4 Uniform coefficient crossover and its Landau limit

4.1 The limiting density and a domination lemma

For $\lambda > 0$, define

$$\mathcal{H}_\varepsilon(\lambda) = \frac{1}{2\pi i} \int_{1-i\infty}^{1+i\infty} \exp\{-\lambda f_\varepsilon(s)\} ds. \quad (4.1)$$

The integral is absolutely convergent. We write $\mathcal{H} = \mathcal{H}_0$. A useful probabilistic normalization is the Lévy measure

$$\nu_{\varepsilon,\lambda}(dx) = \frac{\lambda}{\Gamma(1 - \varepsilon)} e^{-x} x^{-2-\varepsilon} dx, \quad x > 0. \quad (4.2)$$

Although its first moment at zero is infinite, its second moment is finite. Hence compensated integration of the jumps, defined as an L^2 limit, gives a centered random variable $Y_{\varepsilon,\lambda}$ with characteristic exponent

$$\Psi_{\varepsilon,\lambda}(u) = \int_0^\infty (e^{iux} - 1 - iux) \nu_{\varepsilon,\lambda}(dx) \quad (4.3)$$

$$= \lambda \{f_\varepsilon(1) - f_\varepsilon(1 - iu)\}. \quad (4.4)$$

At zero,

$$\Psi_{0,\lambda}(u) = \lambda \{(1 - iu) \log(1 - iu) + iu\}. \quad (4.5)$$

Its density will be denoted by $g_{\varepsilon,\lambda}$.

Lemma 4.1 (Decay, analyticity, and positivity). *For $0 \leq \varepsilon \leq \eta < 1/2$ and λ in a compact positive interval,*

$$|\exp \Psi_{\varepsilon,\lambda}(u)| \leq C \exp(-b|u|^{1+\varepsilon}) \quad (|u| \geq 1),$$

where $C, b > 0$ are uniform. The densities $g_{\varepsilon,\lambda}$ exist, are real analytic, and are strictly positive on $\mathbb{R}$. Moreover

$$g_{\varepsilon,\lambda}(0) = e^{\lambda/(1+\varepsilon)} \mathcal{H}_\varepsilon(\lambda) > 0. \quad (4.6)$$

For each fixed $m > 0$, the same analyticity and positivity statements hold when $\nu_{\varepsilon,\lambda}$ is restricted to $(0, m]$.

Proof. For $|u| \geq 1$, integrate the nonnegative quantity $(1 - \cos ux)e^{-x}x^{-2-\varepsilon}$ over $x \in [a/|u|, b_0/|u|]$, where fixed $0 < a < b_0 < 2\pi$ avoid zeros of $1 - \cos v$. The resulting lower bound is a uniform positive multiple of $|u|^{1+\varepsilon}$. For a restricted measure the same interval works as soon as $|u| \geq b_0/m$. Fourier inversion therefore gives a continuous density, whose Fourier integral extends holomorphically to a strip. It is a nonnegative real-analytic function and is not identically zero; its real zeros are consequently isolated. Splitting the intensity into two equal independent parts expresses this density as the convolution of two such densities. Each factor is positive almost everywhere, so the convolution is positive at every real argument. This proves strict positivity, including in the restricted case. Formula (4.6) follows from (4.4). Equation (4.4) itself may be checked by differentiating twice in u , evaluating the elementary gamma integral, and matching value and first derivative at zero. $\square$

Lemma 4.2 (Uniform lattice attenuation). *Let μ_j be given by (2.7) and set*

$$D(\theta) = \sum_{j \geq 1} \mu_j (1 - \cos j\theta).$$

There exist fixed $\theta_0, b > 0$, uniform for small δ , $0 \leq \varepsilon \leq \eta$, and compact positive λ -windows, such that

$$D(\theta) \geq b\lambda(\theta/\delta)^2, \quad |\theta| \leq \delta, \quad (4.7)$$

$$D(\theta) \geq b\lambda|\theta/\delta|^{1+\varepsilon}, \quad \delta \leq |\theta| \leq \theta_0, \quad (4.8)$$

$$D(\theta) \geq bnc, \quad \theta_0 \leq |\theta| \leq \pi. \quad (4.9)$$

Proof. For $|\theta| \leq \delta$, take integers $j \in [1/(2\delta), 1/\delta]$. There are order $1/\delta$ of them, $e^{-\delta j}$ is bounded below, and $1 - \cos j\theta \geq b_1 j^2 \theta^2$. Their sum is at least $b_2 nc \delta^{\varepsilon-1} \theta^2$, which equals a positive uniform multiple of $\lambda(\theta/\delta)^2$. For $\delta \leq |\theta| \leq \theta_0$, use $j \in [a/|\theta|, b_0/|\theta|]$ with fixed suitable a, b_0 . Choosing θ_0 small ensures enough integers in this interval; the phase is away from a multiple of 2π , and $e^{-\delta j}$ stays bounded below. The sum is at least $b_3 nc |\theta|^{1+\varepsilon}$. Since $nc \delta^{1+\varepsilon} = \lambda/\Gamma(1-\varepsilon)$, this proves the middle bound. For the last bound, the single $j = 1$ term suffices because $1 - \cos \theta$ has a positive minimum on $\theta_0 \leq |\theta| \leq \pi$. $\square$

4.2 A uniform theorem, not merely a formal substitution

Theorem 4.3 (Lower-endpoint coefficient crossover). *Fix $\eta < 1/2$ and $0 < \lambda_- < \lambda_+ < \infty$. Uniformly over $0 \leq \varepsilon \leq \eta$, $\delta \downarrow 0$, and integers $n \geq 1$ for which $\lambda = nA\delta \in [\lambda_-, \lambda_+]$,*

$$u_n(\varepsilon, c) = \rho_0^{-n} \frac{\delta}{n} \left\{ \mathcal{H}_\varepsilon(\lambda) + O(\delta^{1-\eta}) \right\}. \quad (4.10)$$

The error is additive inside braces, and is also a relative $O(\delta^{1-\eta})$ error because the profile is bounded away from zero on these compact parameter sets. If additionally $\varepsilon \rightarrow 0$, then $\mathcal{H}_\varepsilon(\lambda) \rightarrow \mathcal{H}(\lambda)$ uniformly on compact positive λ -intervals. In particular no bound on $\varepsilon \log(1/\delta)$ is required for the joint limit.

Proof. Cauchy's coefficient integral on $|t| = e^{-\delta}$, with $t = e^{-\delta+i\theta}$, is exactly

$$\frac{n\rho_0^n u_n}{\delta} = \frac{1}{2\pi} \int_{-\pi/\delta}^{\pi/\delta} \exp\{-\lambda F_{\varepsilon,\delta}(1 - iu)\} du. \quad (4.11)$$

There is no extra factor $e^{-\delta s}$ in this formula: we are extracting a coefficient of U , and the original contour element is dt/t . Including such a factor would give incorrect correction coefficients.

The value of the integrand's modulus at $u = 0$ is uniformly bounded. Its remaining modulus is multiplied by $\exp\{-D(\delta u)\}$. Lemma 4.2 therefore supplies an integrable dominating function on the central and intermediate arcs; the far arc has integral at most $C\delta^{-1}\exp(-bnc)$, smaller than every fixed positive power of δ . The normal form gives pointwise convergence of the central integrand to $\exp\{-\lambda f_\varepsilon(1 - iu)\}$.

To obtain the stated rate rather than only convergence, fix a small $r > 0$. The convergent series (3.4) gives, on $|\delta(1 - iu)| \leq r$,

$$|R_{\varepsilon,\delta}(1 - iu)| \leq C\delta^{1-\varepsilon}(1 + u^2). \quad (4.12)$$

Both the exact exponent and the limiting exponent have real-part attenuation at least $b|u|^{1+\varepsilon} - C$ for $|u| \geq 1$. Using the integral formula for the difference of two exponentials, their interpolated real parts have the same bound. The difference of the integrands is thus bounded by

$$C\delta^{1-\varepsilon}(1 + u^2)\exp(-b|u|^{1+\varepsilon}).$$

Its integral is $O(\delta^{1-\eta})$. Beyond this growing arc the attenuation estimates make both tails exponentially small in a positive power of $1/\delta$. This proves (4.10).

Finally, $f_\varepsilon(s)$ is continuous at zero in ε , and the same uniform domination applies to (4.1). Dominated convergence gives continuity of the profile in (ε, λ) . Strict positivity from Lemma 4.1, followed by compactness, gives the positive lower bound and the claimed relative interpretation. $\square$

Corollary 4.4 (The essential crossover scale). *At $\varepsilon = 0$, the bounded positive crossover occurs at*

$$\lambda = nc\delta \sim nce^{-1/c}, \quad n \asymp \frac{e^{1/c}}{c}. \quad (4.13)$$

Thus an exponentially close fold becomes visible only at an exponentially large coefficient index. This is an index scale, not an additional singularity of the original equation.

Proof. Combine (2.5) with $A = c$ and the definition of λ . $\square$

5 The Landau profile and an all-orders correction calculus

5.1 Identification with a classical density

Define the Landau density in the following normalization:

$$p(x) = \frac{1}{2\pi i} \int_{a-i\infty}^{a+i\infty} \exp\{z \log z + xz\} dz, \quad a > 0. \quad (5.1)$$

The value is independent of a , by contour displacement and the exponential decay on vertical lines. Different location and scale conventions for this classical law are common [5]; equation (5.1), rather than a name alone, fixes our convention.

Proposition 5.1 (Exact Landau identification). *For every $\lambda > 0$,*

$$\mathcal{H}(\lambda) = \frac{1}{\lambda} p(-1 - \log \lambda), \quad (5.2)$$

$$g_{0,\lambda}(y) = \frac{e^{\lambda-y}}{\lambda} p\left(\frac{y}{\lambda} - 1 - \log \lambda\right). \quad (5.3)$$

The function p is strictly positive and integrates to one. It also has the real Hankel representation

$$p(x) = \frac{1}{\pi} \int_0^\infty e^{-t \log t - xt} \sin(\pi t) dt. \quad (5.4)$$

Proof. In (4.1) at $\varepsilon = 0$, substitute $z = \lambda s$. The exponent becomes $z \log z + (-1 - \log \lambda)z$, proving (5.2). Fourier inversion of (4.5), with $s = 1 - iu$ and then $z = \lambda s$, gives (5.3). Positivity follows from Lemma 4.1. To check normalization without relying on a distributional naming convention, use $\lambda = 1$: $p(x) = e^x g_{0,1}(x + 1)$. The exponential moment at one exists and

$$\log \mathbb{E} e^{Y_{0,1}} = \int_0^\infty \frac{1 - (1+x)e^{-x}}{x^2} dx = 1,$$

where integration by parts evaluates the integral. Thus $\int p(x) dx = e^{-1} \mathbb{E} e^{Y_{0,1}} = 1$. Finally deform the vertical contour to the two sides of the negative real axis. On those sides $z \log z = -t \log t \mp i\pi t$. Their jump gives (5.4). The small circle about zero contributes nothing, and the large arcs vanish because of the $z \log z$ decay in the left half-plane. The resulting real integral is absolutely convergent for every fixed real x . $\square$

5.2 The two tails of the coefficient profile

Theorem 5.2 (Landau profile tails). *As $\lambda \rightarrow \infty$,*

$$\mathcal{H}(\lambda) = \frac{e^{-\lambda}}{\sqrt{2\pi\lambda}} \left(1 + \frac{1}{24\lambda} - \frac{23}{1152\lambda^2} + O(\lambda^{-3}) \right), \quad (5.5)$$

and an expansion to every fixed inverse power of λ exists. As $\lambda \downarrow 0$, putting $x = -1 - \log \lambda$,

$$\mathcal{H}(\lambda) = \frac{1}{\lambda x^2} \left(1 + O\left(\frac{\log x}{x}\right) \right). \quad (5.6)$$

Equivalently, the Landau left tail satisfies, for $h = e^{-1-x} \rightarrow \infty$,

$$p(x) = \sqrt{\frac{h}{2\pi}} e^{-h} \left(1 + \frac{1}{24h} + O(h^{-2}) \right). \quad (5.7)$$

Proof. Write $\phi(s) = s \log s - s$. On $s = 1 + iu$,

$$\operatorname{Re}(\phi(1 + iu) + 1) = \frac{1}{2} \log(1 + u^2) - u \arctan u.$$

Its derivative for $u > 0$ is $-\arctan u$. It is therefore at most $-b \min(u^2, |u|)$, for a fixed $b > 0$. This gives a complete saddle-localization bound, not just a formal Taylor expansion. With $u = v/\sqrt{\lambda}$, the exponent is

$$\lambda \phi(1 + iv/\sqrt{\lambda}) = -\lambda - \frac{v^2}{2} + \frac{iv^3}{6\sqrt{\lambda}} + \frac{v^4}{12\lambda} - \frac{iv^5}{20\lambda^{3/2}} - \frac{v^6}{30\lambda^2} + \dots$$

Taylor expansion on a growing central interval, followed by the above tail bound, justifies termwise Gaussian integration to any fixed order. The coefficient of λ^{-1} is $\mathbb{E} Z^4/12 - \mathbb{E} Z^6/72 = 1/24$, where Z is standard normal. The next coefficient is

$$-\frac{15}{30} + 105 \left(\frac{1}{120} + \frac{1}{288} \right) - \frac{945}{864} + \frac{10395}{31104} = -\frac{23}{1152}.$$

This proves (5.5); (5.7) follows from (5.2) with $\lambda = h$.

For the other tail, substitute $t = v/x$ in (5.4). The leading integrand is $e^{-v} \pi v/x$; after the outer factor $1/(\pi x)$, its integral is x^{-2} . On $v \leq \sqrt{x}$, expanding $\exp\{-(v/x) \log(v/x)\}$ and $\sin(\pi v/x)$ gives an integrated error $O(x^{-3} \log x)$. On the complement, use $-t \log t \leq 1/e$ to bound the unexpanded integrand by an exponentially small tail in $\sqrt{x}$. Thus $p(x) = x^{-2} + O(x^{-3} \log x)$, and (5.6) follows. $\square$

5.3 Finite-coefficient corrections of every order

At $\varepsilon = 0$, write the convergent normal-form remainder as

$$R_{0,\delta}(s) = \sum_{j \geq 1} \delta^j r_j(s), \quad r_j(s) = \frac{(-1)^{j+1} \zeta(1-j)}{(j+1)!} ((j+1)s - s^{j+1}). \quad (5.8)$$

Define finite polynomials in s and λ by

$$P_r(s; \lambda) = [d^r] \exp \left\{ -\lambda \sum_{j \geq 1} d^j r_j(s) \right\}, \quad (5.9)$$

$$C_r(\lambda) = \frac{1}{2\pi i} \int_{1-i\infty}^{1+i\infty} P_r(s; \lambda) e^{-\lambda f_0(s)} ds. \quad (5.10)$$

In (5.9), only $j \leq r$ can contribute.

Theorem 5.3 (All-orders endpoint coefficient expansion). *For every integer $N \geq 1$ and every compact $K \subset (0, \infty)$, uniformly when $\lambda = nc\delta \in K$,*

$$\frac{n\rho_0^n u_n(0, c)}{\delta} = \sum_{r=0}^{N-1} \delta^r C_r(\lambda) + O_{K,N}(\delta^N). \quad (5.11)$$

Here $C_0 = \mathcal{H}$, $\deg_s P_r \leq 2r$, and the polynomials satisfy

$$P_0 = 1, \quad rP_r = -\lambda \sum_{j=1}^r j r_j P_{r-j}. \quad (5.12)$$

If $P_r(s; \lambda) = \sum_m a_{r,m}(\lambda) s^m$, then

$$C_r(\lambda) = \sum_{m=0}^{2r} a_{r,m}(\lambda) \lambda^{-m-1} p^{(m)}(-1 - \log \lambda). \quad (5.13)$$

This is an asymptotic expansion to arbitrary finite order; convergence of the infinite r -sum is not asserted.

Proof. Differentiate the generating exponential in (5.9) with respect to d . Coefficient comparison proves (5.12). Since $\deg r_j = j+1 \leq 2j$, every product of total d -degree r has s -degree at most $2r$.

It remains to justify integration of the expansion. The exact contour formula is (4.11). On a growing central arc $|u| \leq r/\delta$, with $r > 0$ sufficiently small, the convergent series for $R_{0,d}(1 - iu)$ is analytic for real $0 \leq d \leq \delta$. For each fixed N , its first N derivatives in d are bounded by fixed polynomials in $1 + |u|$, uniformly on this arc. Moreover

$$\operatorname{Re}\{f_0(1 - iu) + R_{0,d}(1 - iu)\} \geq b|u| - C$$

there, uniformly in d : the leading attenuation is linear at large $|u|$, whereas $|R_{0,d}| \leq Cd(1 + u^2)$, and $d|u| \leq r$. Choose r small enough to absorb this last contribution. Taylor's theorem with an integral remainder, applied to the exponential as a function of d , now bounds the remainder by $C_N \delta^N (1 + |u|)^{B_N} e^{-b'|u|}$. It is integrable uniformly in δ . The exact and polynomial-model tails outside the central arc are exponentially small by Lemma 4.2 and the explicit limiting attenuation. Thus termwise integration gives (5.11). Finally substituting $z = \lambda s$ into the integral of s^m , and differentiating (5.1) under its absolutely convergent integral, gives (5.13). $\square$

Example 5.4 (First two corrections). The first two coefficient polynomials are

$$P_1(s; \lambda) = \frac{\lambda}{2} \left(s - \frac{s^2}{2} \right), \quad (5.14)$$

$$P_2(s; \lambda) = -\frac{\lambda}{72}(3s - s^3) + \frac{\lambda^2}{8} \left(s - \frac{s^2}{2} \right)^2. \quad (5.15)$$

For $x = -1 - \log \lambda$, in particular,

$$C_1(\lambda) = \frac{p'(x)}{2\lambda} - \frac{p''(x)}{4\lambda^2}. \quad (5.16)$$

Equivalently, with $I_m = \lambda^{-m-1}p^{(m)}(x)$,

$$C_1 = \frac{\lambda}{2}(I_1 - I_2/2), \quad C_2 = -\frac{\lambda}{72}(3I_1 - I_3) + \frac{\lambda^2}{8}(I_2 - I_3 + I_4/4).$$

Every further correction is generated by the finite recurrence (5.12); no fresh inversion or saddle equation is required.

6 Convergent inverse transseries and the moving two-sheet chart

For this section $\varepsilon = 0$. Normalize the inverse variable by

$$w = \frac{\log(q/\rho_0)}{c\delta}. \quad (6.1)$$

The equation for $s = x/\delta$ is simply $w = F_{0,\delta}(s)$. At $\delta = 0$ it becomes $w = s(1 - \log s)$, whose physical real solution for $w < 1$ is

$$S_0(w) = \exp\{1 + W_0(-w/e)\} > 1. \quad (6.2)$$

Here W_0 is the principal Lambert branch, characterized by $We^W = z$ and continued from zero [6].

Theorem 6.1 (Convergent exponential-scale inverse). *For every compact real interval $I \subset (-\infty, 1)$, there is a complex neighborhood of I and an $r > 0$ on which the physical solution has a convergent expansion*

$$S(w, \delta) = S_0(w) + \sum_{k \geq 1} \delta^k S_k(w), \quad |\delta| < r. \quad (6.3)$$

It satisfies

$$S_1(w) = \frac{S_0(w)^2/4 - S_0(w)/2}{\log S_0(w)}. \quad (6.4)$$

On every smaller product domain, the series has a geometric remainder bound. In the original variables,

$$U(q) = -\log q - \delta S(w, \delta), \quad \delta = -\log(1 - E), \quad E = e^{-1/c}. \quad (6.5)$$

Thus, at fixed normalized w , (6.5) yields a genuinely convergent expansion in the exponential monomial E .

Proof. The endpoint remainder (3.4) is jointly holomorphic in (s, δ) near every point $(S_0(w), 0)$ with $w \in I$. The derivative of the leading equation in s is $-\log S_0(w)$, uniformly separated from zero on I . The analytic implicit function theorem supplies local holomorphic solutions. Their uniqueness makes them agree on overlaps; a finite cover of the compact interval gives a common positive δ -radius and a complex neighborhood of I . Taylor expansion in δ proves (6.3). Comparing its first-order term gives

$$-\log S_0 S_1 + (-S_0/2 + S_0^2/4) = 0,$$

which is (6.4). If the solution is bounded by B on $|\delta| \leq r_0$, Cauchy's estimate gives, for $|\delta| < r_0$,

$$\left| S - \sum_{k=0}^N \delta^k S_k \right| \leq B \frac{(|\delta|/r_0)^{N+1}}{1 - |\delta|/r_0}.$$

The identity $t = qe^U = e^{-\delta S}$ proves (6.5). Finally $-\log(1 - E)$ is analytic at $E = 0$, so substitution preserves convergence on a smaller disk. More explicitly, the original expansion has the form $U = c\zeta(2) + \sum_{k \geq 1} (a_k(w) + cb_k(w))E^k$, with coefficient functions independent of c . Convergence in E is not analyticity in c at zero. This is a normalized chart: it does not assert convergence for a fixed unscaled q as $c \downarrow 0$. $\square$

Theorem 6.2 (Uniform two-sheet chart through the fold). *Put*

$$w_c(\delta) = F_{0,\delta}(1), \quad a(\delta) = \frac{\delta}{e^\delta - 1}, \quad \kappa = w_c(\delta) - w. \quad (6.6)$$

There is a function $\chi(\delta, z)$, jointly holomorphic near $(0, 0)$, with $\chi(\delta, 0) = 0$ and $\partial_z \chi(\delta, 0) = 1$, such that the two inverse sheets are

$$s = 1 + \chi \left(\delta, \pm \sqrt{\frac{2\kappa}{a(\delta)}} \right). \quad (6.7)$$

Its double Taylor expansion has geometric remainder bounds on smaller polydisks. For positive real κ , the plus sign gives the physical branch $s > 1$.

Proof. By exact stationarity and (3.9),

$$B(\delta, s) = \frac{2(F_{0,\delta}(1) - F_{0,\delta}(s))}{a(\delta)(s - 1)^2}$$

extends holomorphically across $s = 1$, with $B(\delta, 1) = 1$. Choose its holomorphic square root near this value. The map $z = (s - 1)\sqrt{B(\delta, s)}$ has derivative one in s at the fold, so it has a jointly holomorphic inverse $s = 1 + \chi(\delta, z)$. The identity $z^2 = 2\kappa/a(\delta)$ proves (6.7). Cauchy estimates on a smaller polydisk give the remainder bounds. The sign assertion follows by continuity from $s - 1 = z + O(z^2)$. $\square$

Remark 6.3 (Two different kinds of series). The convergent inverse series of Theorems 6.1 and 6.2 should not be confused with the all-orders large-index expansion in Theorem 5.3. The former is a local analytic statement in a normalized inverse coordinate; the latter integrates correction polynomials over an unbounded contour. Analyticity of the local chart alone does not prove convergence of that integrated asymptotic expansion.

7 Gaussian matching and the first nonuniformity threshold

A compact-window crossover formula cannot automatically be extrapolated to $\lambda \rightarrow \infty$. The following direct proof both supplies the Gaussian regime and quantifies the first obstruction to doing so.

Theorem 7.1 (Uniform endpoint Gaussian regime). *Let $\varepsilon = 0$, $\delta \rightarrow 0$, and $\lambda = nc\delta \rightarrow \infty$, with c still given by (2.1). Then*

$$u_n(0, c) = \frac{\rho^{-n}}{n\sqrt{2\pi nc \operatorname{Li}_0(e^{-\delta})}}(1 + o(1)) \quad (7.1)$$

$$= \rho_0^{-n} \frac{\delta}{n\sqrt{2\pi\lambda}} \sqrt{\frac{e^\delta - 1}{\delta}} \exp\{-\lambda F_{0,\delta}(1)\}(1 + o(1)). \quad (7.2)$$

No assumption on $\lambda\delta$ is needed when the exact fold value is retained.

Proof. In the tilted Poisson representation,

$$\sigma_n^2 = \operatorname{Var} S = nc \operatorname{Li}_0(e^{-\delta}) \sim \frac{\lambda}{\delta^2}.$$

For every fixed $r \geq 2$, the r -th cumulant is $nc \operatorname{Li}_{2-r}(e^{-\delta})$. The elementary comparison of a sum with its integral gives $\operatorname{Li}_{2-r}(e^{-\delta}) \sim \Gamma(r-1)\delta^{1-r}$. Thus the standardized cumulants of order $r \geq 3$ are $O(\lambda^{1-r/2})$, and the characteristic function on $\theta = v/\sigma_n$ tends locally uniformly to $e^{-v^2/2}$. One can also obtain this directly by Taylor expansion with the absolute third moment bound.

The proof of Lemma 4.2 has constants independent of λ ; only its explicit factors of λ change. On the central arc its attenuation becomes bv^2 . On the intermediate arc it is at least a positive multiple of $\sqrt{\lambda}|v|$, and the far-arc integral, after scaling by σ_n , is bounded by $C\sigma_n e^{-bnc} \rightarrow 0$. Fourier inversion and dominated convergence therefore give the lattice local limit $\sigma_n \mathbb{P}(S = n) \rightarrow 1/\sqrt{2\pi}$. Equation (2.8) proves (7.1). Using $\operatorname{Li}_0(e^{-\delta}) = (e^\delta - 1)^{-1}$ and $n \log(\rho/\rho_0) = \lambda F_{0,\delta}(1)$ gives the second form. $\square$

Editorial note (ProveIt, 2026-09-29). For $\varepsilon > 0$ the model (1.1) is the critical-Hahn article's $U = \beta A(qe^U)$ [1] with $a = 1$, no finite prefix, $\beta = c$ and $\alpha = 1 + \varepsilon$; there $A(t) = \operatorname{Li}_{2+\varepsilon}(t)$ is the action generating function, not the amplitude A of (2.2). Its $\beta_c = 1/d_1$ is $c_b = 1/\zeta(1 + \varepsilon)$, and its $\rho_\beta = e^{-\beta d_0}$ is ρ_0 . Its $\delta_\beta = 1 - \beta d_1$ is a coupling mismatch, not the fold distance δ used here. The exact fold of Proposition 2.1 reduces, as $\delta \rightarrow 0$ at fixed ε , to that article's emerging-fold law `eq:emerging-x`. The leading form (7.1), read at fixed c , has the shape of its supercritical law `eq:supercritical` (Theorem `thm:phases`) with $r_\beta = \rho$ and $v_\beta = c \operatorname{Li}_\varepsilon(\tau)$, here $c \operatorname{Li}_0(e^{-\delta})$; that law is stated there for $1 < \alpha < 2$ only. Its cutoff criterion $M_n/n^{1/\alpha} \rightarrow \infty$ is at the boundary-critical coupling, whereas $M\delta \rightarrow \infty$ of Corollary 8.2 is on the interior-fold side: different regimes, no conflict. This comparison is editorial.

Corollary 7.2 (The first matching obstruction). *Under the hypotheses of Theorem 7.1,*

$$\frac{u_n(0, c)}{\rho_0^{-n}(\delta/n)\mathcal{H}(\lambda)} = \sqrt{\frac{e^\delta - 1}{\delta}} \exp\{\lambda(1 - F_{0,\delta}(1))\}(1 + o(1)). \quad (7.3)$$

If $\lambda\delta \rightarrow \ell \in [0, \infty)$, this ratio tends to $e^{\ell/4}$. If $\lambda\delta \rightarrow \infty$, then

$$\log \frac{u_n(0, c)}{\rho_0^{-n}(\delta/n)\mathcal{H}(\lambda)} \sim \frac{\lambda\delta}{4}. \quad (7.4)$$

Thus the leading Landau formula matches the unbounded Gaussian regime without a relative correction precisely on the additional overlap scale $\lambda\delta \rightarrow 0$.

Proof. Divide (7.2) by (5.5). The factor in parentheses in (5.5) tends to one. By (3.8),

$$\lambda(1 - F_{0,\delta}(1)) = \lambda\delta/4 - \lambda\delta^2/36 + O(\lambda\delta^4).$$

The remaining square root tends to one. This proves all assertions, including the logarithmic equivalent when $\lambda\delta$ diverges. For the final “precisely” statement, a positive finite or infinite subsequential limit of $\lambda\delta$ prevents the ratio from tending to one; the preceding formulas cover each such subsequence. $\square$

Remark 7.3 (Why the exact normalization matters). The factor $e^{\lambda\delta/4}$ is not a higher-order change in a bounded prefactor once λ grows. It is an exponential drift of the fold value. Keeping ρ exact avoids accumulating this error. This is a concrete instance of a general warning in asymptotic inversion: a small error in a singularity location can become a leading error after multiplication by a large coefficient index.

8 Conditional action cutoffs at the lower endpoint

For the remainder of the limiting-profile discussion, abbreviate

$$\nu_\lambda(dx) = \lambda e^{-x} x^{-2} dx, \quad Y_\lambda = Y_{0,\lambda}, \quad g_\lambda = g_{0,\lambda}.$$

For $m > 0$, let $Y_{\lambda,m}$ be the compensated jump variable with measure restricted to $(0, m]$, and let $g_{\lambda,m}$ be its density. Define

$$b_m = \int_m^\infty \nu_\lambda(dx) = \lambda \left(\frac{e^{-m}}{m} - E_1(m) \right), \quad (8.1)$$

$$a_m = \int_m^\infty x \nu_\lambda(dx) = \lambda E_1(m), \quad (8.2)$$

$$\mathcal{R}(\lambda, m) = \frac{e^{-b_m} g_{\lambda,m}(a_m)}{g_\lambda(0)}, \quad (8.3)$$

where $E_1(z) = \int_1^\infty e^{-zt} dt/t$ for $\operatorname{Re} z > 0$. The sign of a_m in (8.3) is positive. Removing all large jumps leaves the negative compensation $-a_m$, so the small-jump variable must equal $+a_m$ to bring the full centered sum to zero.

Theorem 8.1 (Joint lower-endpoint cutoff profile). *Let $\varepsilon \rightarrow 0$, $\delta \rightarrow 0$, $\lambda = nA\delta \rightarrow \lambda_0 \in (0, \infty)$, and $M\delta \rightarrow m \in (0, \infty)$. Then*

$$\frac{u_{n,M}(\varepsilon, c)}{u_n(\varepsilon, c)} \longrightarrow \mathcal{R}(\lambda_0, m). \quad (8.4)$$

Convergence is uniform when λ and $M\delta$ range over compact positive intervals and $\varepsilon \rightarrow 0$. No condition on $\varepsilon \log(1/\delta)$ is imposed. For fixed $\lambda > 0$, $\mathcal{R}(\lambda, m)$ is continuous, strictly increasing in m , lies strictly between zero and one, and satisfies

$$\lim_{m \downarrow 0} \mathcal{R}(\lambda, m) = 0, \quad \lim_{m \rightarrow \infty} \mathcal{R}(\lambda, m) = 1. \quad (8.5)$$

Proof. The characteristic function of $\delta(S - n)$ in the exact Poisson model converges to $\exp \Psi_{0,\lambda_0}$, by the normal form or directly by compensated Riemann sums. Indeed the intensity at $x = \delta j$, divided by its mesh δ , is

$$\frac{\mu_j}{\delta} = \frac{\lambda}{\Gamma(1 - \varepsilon)} e^{-x} x^{-2 - \varepsilon}.$$

Near zero the compensated integrand is $O(x^2)$, uniformly for bounded Fourier arguments; away from zero the exponential tail dominates. These estimates also prove uniformity on the stated compact parameter sets.

Imposing no jump above M gives the killed characteristic function

$$\mathbb{E} \left[e^{iu\delta(S-n)} \mathbf{1}_{\{N_j=0 \ (j>M)\}} \right] = \mathbb{E} e^{iu\delta(S-n)} \exp \left\{ - \sum_{j>M} \mu_j e^{iu\delta j} \right\}. \quad (8.6)$$

On compact positive $M\delta$ -windows the total omitted intensity is uniformly bounded. Thus the extra exponential in (8.6) is bounded in modulus by a fixed constant, and the Fourier domination of Lemma 4.2 still applies. Fourier inversion at zero and dominated convergence give the joint local limit

$$\delta^{-1} \mathbb{P}(S = n, N_j = 0 \text{ for } j > M) \longrightarrow \frac{1}{2\pi} \int_{\mathbb{R}} \exp \left\{ \Psi_{0,\lambda_0}(u) - \int_m^\infty e^{iux} \nu_{\lambda_0}(dx) \right\} du.$$

Split the limiting variable as $Y_\lambda = Y_{\lambda,m} + J_{>m} - a_m$, where $J_{>m}$ is the uncompensated finite Poisson sum of jumps above m . The displayed integral is exactly $e^{-b_m} g_{\lambda,m}(a_m)$. The denominator local limit is $\delta^{-1} \mathbb{P}(S = n) \rightarrow g_\lambda(0) > 0$. Equation (2.9) proves (8.4).

Positivity of $g_{\lambda,m}$ proves $\mathcal{R} > 0$. The event of exactly one jump in any fixed interval above m , no other jump above m , and the required small-jump value has positive density at zero. Hence $\mathcal{R} < 1$. The same argument with an interval contained in $(m_1, m_2]$ shows strict increase between any two cutoffs. Continuity follows either from the killed Fourier integral or from Poisson splitting and dominated convergence. As $m \rightarrow \infty$, the omitted intensity tends to zero and that Fourier integral tends to the unrestricted density, proving the second limit in (8.5).

For the small-cutoff limit, the real part of the restricted characteristic exponent has the elementary lower attenuation bounds $b\lambda mu^2$ when $|u| \leq 1/m$, and $b\lambda|u|$ when $|u| \geq 1/m$, for small m . The proof is the interval argument of Lemma 4.1, using the whole interval $(0, m]$ in the first case. Thus $\sup_y g_{\lambda,m}(y) = O_\lambda(m^{-1/2})$. But $b_m \sim \lambda/m$, so the factor e^{-b_m} in (8.3) overwhelms this polynomial bound. This proves $\mathcal{R} \rightarrow 0$ and completes the proof. $\square$

Corollary 8.2 (Sharp coefficient-preserving action budget). *Suppose $\varepsilon \rightarrow 0$, $\delta \rightarrow 0$, and $\lambda = nA\delta$ stays in a compact subset of $(0, \infty)$. Then, for any positive integer cutoff sequence M ,*

$$\frac{u_{n,M}}{u_n} \longrightarrow 1 \quad \Longleftrightarrow \quad M\delta \longrightarrow \infty. \quad (8.7)$$

At $\varepsilon = 0$, this is equivalent to $Me^{-1/c} \rightarrow \infty$. If instead $M\delta \rightarrow 0$, the retained coefficient fraction tends to zero.

Proof. For sufficiency, compare with the cutoff $\lfloor m/\delta \rfloor$ at any fixed $m > 0$, which is eventually smaller than M . By monotonicity and Theorem 8.1, every subsequential limiting retained fraction is at least $\mathcal{R}(\lambda_0, m)$, after extracting a convergent subsequence of λ . Let $m \rightarrow \infty$. This yields one. If $M\delta$ does not tend to infinity, it has a bounded subsequence. Extract further limits $\lambda \rightarrow \lambda_0 > 0$ and $M\delta \rightarrow m \geq 0$. For $m > 0$, the limit is strictly less than one. For $m = 0$, compare above with every fixed positive scaled cutoff, then let that cutoff tend to zero, obtaining limit zero. This proves necessity and the last assertion. Equation (2.5) proves the endpoint equivalent. $\square$

8.1 A directly evaluable Fourier formula

Let

$$T(z, m) = \int_m^\infty \frac{e^{-zx}}{x^2} dx = \frac{e^{-zm}}{m} - z E_1(zm), \quad \operatorname{Re} z > 0. \quad (8.8)$$

Integration by parts proves the equality. The preceding proof gives

$$\mathcal{R}(\lambda, m) = \frac{1}{2\pi g_\lambda(0)} \int_{\mathbb{R}} \exp \{ \lambda((1-iu) \log(1-iu) + iu - T(1-iu, m)) \} du. \quad (8.9)$$

This single integral is useful for moderate cutoffs. It is not a good way to subtract $\mathcal{R}$ from one when the loss is far below working precision. The positive Palm integral in the next section is preferable in that regime.

9 A sharp doubly exponential law for the cutoff loss

The event in (8.3) can be interpreted as a conditional no-large-jump probability given $Y_\lambda = 0$. Since this event has probability zero, the conditioning is defined by joint densities:

$$\mathbb{P}(B \mid Y_\lambda = 0) := \frac{\text{density at zero of } \mathbb{P}(Y_\lambda \in dy, B)}{g_\lambda(0)}.$$

All densities used here are continuous, and $g_\lambda(0) > 0$, so this specifies the relevant bridge probabilities without any ambiguity about a version on a null event.

Lemma 9.1 (Density-level Poisson insertion). *Let K_m be the number of jumps of Y_λ larger than m . Its first two conditional factorial moments are*

$$J_1(\lambda, m) := \mathbb{E}(K_m \mid Y_\lambda = 0) = \frac{1}{g_\lambda(0)} \int_m^\infty g_\lambda(-x) \nu_\lambda(dx), \quad (9.1)$$

$$\begin{aligned} J_2(\lambda, m) &:= \mathbb{E}(K_m(K_m - 1) \mid Y_\lambda = 0) \\ &= \frac{1}{g_\lambda(0)} \int_m^\infty \int_m^\infty g_\lambda(-x - y) \nu_\lambda(dx) \nu_\lambda(dy). \end{aligned} \quad (9.2)$$

Consequently,

$$J_1 - \frac{1}{2} J_2 \leq 1 - \mathcal{R}(\lambda, m) \leq J_1. \quad (9.3)$$

In the Landau normalization these integrals simplify to

$$J_1(\lambda, m) = \frac{1}{\mathcal{H}(\lambda)} \int_m^\infty \frac{p(-x/\lambda - 1 - \log \lambda)}{x^2} dx, \quad (9.4)$$

$$J_2(\lambda, m) = \frac{\lambda}{\mathcal{H}(\lambda)} \int_m^\infty \int_m^\infty \frac{p(-(x+y)/\lambda - 1 - \log \lambda)}{x^2 y^2} dx dy. \quad (9.5)$$

Proof. The jumps above m form a finite Poisson process. Summing its Poisson probabilities after marking one point gives, for a bounded test function h ,

$$\mathbb{E}(K_m h(Y_\lambda)) = \int_m^\infty \mathbb{E}h(Y_\lambda + x) \nu_\lambda(dx).$$

The compensator is unchanged when a point is inserted; the full variable is shifted by $+x$. Marking an ordered pair gives the analogous identity with $K_m(K_m - 1)$ and shift $x + y$. These are the usual Poisson insertion identities, here proved directly for the finite marked part of the process. Approximate a point mass at zero by bounded smooth test functions and use the continuous bounded density to obtain (9.1)–(9.2). The pointwise inequalities $k - k(k - 1)/2 \leq \mathbf{1}_{\{k \geq 1\}} \leq k$, for integers $k \geq 0$, give (9.3). Substitute (5.3) and $g_\lambda(0) = e^\lambda \mathcal{H}(\lambda)$. The factors e^{-x} in the jump intensity cancel the corresponding exponential tilt in $g_\lambda(-x)$, leaving exactly (9.4) and (9.5). $\square$

Theorem 9.2 (Doubly exponential cutoff loss with sharp prefactor). *For every fixed $\lambda > 0$, as $m \rightarrow \infty$,*

$$1 - \mathcal{R}(\lambda, m) \sim \frac{\sqrt{\lambda/(2\pi)}}{\mathcal{H}(\lambda)m^2} \exp \left\{ -\frac{m}{2\lambda} - \lambda e^{m/\lambda} \right\}. \quad (9.6)$$

Thus the conditional cutoff loss is not asymptotic to the unconditioned omitted intensity $b_m \sim \lambda e^{-m}/m^2$.

Proof. For $x \rightarrow \infty$, Theorem 5.2 gives

$$p(-x/\lambda - 1 - \log \lambda) = \sqrt{\frac{\lambda}{2\pi}} \exp \left\{ \frac{x}{2\lambda} - \lambda e^{x/\lambda} \right\} \{1 + O_\lambda(e^{-x/\lambda})\}. \quad (9.7)$$

The estimate is uniform for all $x \geq m$ when m is large. Insert it in (9.4) and use the endpoint substitution $x = m + e^{-m/\lambda}t$. After dividing by the proposed right-hand side of (9.6), the integral becomes

$$\int_0^\infty \left(\frac{m}{m + e^{-m/\lambda}t} \right)^2 \exp \left\{ \frac{e^{-m/\lambda}t}{2\lambda} - \lambda e^{m/\lambda} \left(\exp \left(\frac{e^{-m/\lambda}t}{\lambda} \right) - 1 \right) \right\} \{1 + O_\lambda(e^{-m/\lambda})\} dt.$$

Its integrand tends to e^{-t} . The inequality $e^v - 1 \geq v$ bounds the exponential by $e^{-t/2}$ for all large m , uniformly for $t \geq 0$. The other factors are bounded. Dominated convergence shows that this integral tends to one. Therefore J_1 has the equivalent in (9.6).

It remains to pass from the expected number of large jumps to the probability of at least one. Put $h_v = \lambda e^{v/\lambda}$. The left-tail bound in (5.7), and the fact that $\sqrt{h}e^{-h}$ decreases for large h , imply

$$J_2(\lambda, m) \leq C_\lambda m^{-2} \sqrt{h_{2m}} e^{-h_{2m}}.$$

Indeed, in (9.5), $x + y \geq 2m$, and $\int_m^\infty x^{-2} dx = 1/m$. Comparing this estimate with the just-proved equivalent for J_1 gives $J_2/J_1 \rightarrow 0$: its logarithm is bounded above by $-\lambda e^{2m/\lambda} + \lambda e^{m/\lambda} + O_\lambda(m)$. The two-sided inequality (9.3) now proves (9.6). Finally (8.1), or one elementary integration by parts, yields the stated unconditioned tail equivalent. $\square$

Corollary 9.3 (Double-logarithmic tolerance budget). *For $0 < h < 1$, let m_h be the unique solution of $1 - \mathcal{R}(\lambda, m_h) = h$. For fixed $\lambda > 0$,*

$$m_h = \lambda \log \log(1/h) - \lambda \log \lambda + o(1), \quad h \downarrow 0. \quad (9.8)$$

The more informative implicit asymptotic relation is

$$\log(1/h) = \lambda e^{m_h/\lambda} + \frac{m_h}{2\lambda} + 2 \log m_h + \log \left(\mathcal{H}(\lambda) \sqrt{\frac{2\pi}{\lambda}} \right) + o(1). \quad (9.9)$$

Proof. Existence and uniqueness follow from strict monotonicity and the limits in (8.5). Taking logarithms of (9.6) proves (9.9). Its exponential term dominates the linear and logarithmic terms. Inverting the leading term gives (9.8); the additive error in m_h is $o(1)$, because the derivative of $\lambda e^{m/\lambda}$ tends to infinity much faster than the omitted terms. $\square$

9.1 Interpretation and a necessary qualification

In the endpoint crossover $n \sim \lambda/(c\delta)$, retaining a fixed limiting fraction of a coefficient takes $M \sim m/\delta$ actions. Thus $M/n \sim mc/\lambda \rightarrow 0$ even though M itself is exponentially large in $1/c$. For a subsequently small tolerance, the leading limiting budget is

$$M_h \sim \delta^{-1} \lambda \log \log(1/h)$$

when $h \downarrow 0$ after passing to the profile and with fixed λ . Formula (9.8) retains the next constant shift.

The order of limits matters. Theorem 8.1 is uniform for m in compact positive intervals; it does not supply an error bound uniform for $m = m_n \rightarrow \infty$ at a prescribed finite- n tolerance. Consequently (9.8) must not be marketed as a certified finite computation rule with no remainder estimate. Establishing that uniform two-parameter error bound is an explicit further research problem below. The distinction is particularly important here, because the limiting loss becomes much smaller than ordinary rounding errors.

10 Reproducible checks and worked numerical examples

The proofs above do not depend on numerical evidence. The accompanying program audits signs, normalizations, finite identities, and the onset of the stated asymptotic regimes. The recorded run used Python 3.13.5, SymPy 1.14.0, mpmath 1.3.0, NumPy 2.3.5, and SciPy 1.17.0. No floating-point value below is an interval certificate.

10.1 Exact algebra and high-precision normal forms

For $c = 1/3, 1/2, 2$, the verification program independently computes u_n , $1 \leq n \leq 12$, in two ways. One is direct formal iteration of (1.1); the other is the Lagrange exponential coefficient in (2.6). All 36 rational comparisons agree. Seven additional symbolic assertions check P_1 , P_2 , the first inverse-chart correction, fold stationarity, fold curvature through order seven, and the first two fold-value coefficients. All 43 exact assertions pass. Separately, 24 evaluations at 90-digit working precision compare the exact normal form with its sum through $k = 18$, for $\varepsilon = 0, 0.001, 0.03, 0.2$, two fold distances, and three real or complex s -values. Every absolute discrepancy is below 10^{-39} . Those last comparisons use arbitrary-precision floating arithmetic, not exact or interval arithmetic.

10.2 Coefficient recurrence and correction accuracy

For each target n , put $p_k = \mathbb{P}(S = k)$ in the tilted endpoint model. The positive recurrence

$$p_0 = \exp\{-nc \operatorname{Li}_2(\tau)\}, \quad kp_k = nc \sum_{j=1}^k \frac{\tau^j}{j} p_{k-j} \quad (10.1)$$

follows by differentiating its probability generating function. It computes the target mass without subtractive cancellation. The code uses extended-range `longdouble` for this recurrence and explicitly rejects a platform on which its initial probability underflows. Exact rational checks use a separate implementation.

For $\lambda = 0.25, 1, 4$ and $n = 128, 512, 2048$, the code solves $nc\delta = \lambda$ with the exact fold coupling, then compares

$$Z_n = \frac{n\rho_0^n u_n}{\delta} \quad \text{with} \quad \mathcal{H}, \quad \mathcal{H} + \delta C_1, \quad \mathcal{H} + \delta C_1 + \delta^2 C_2.$$

The following signed relative residuals are divided by $\mathcal{H}$.

λ	n	Leading only	Through δC_1	Through $\delta^2 C_2$
0.25	128	$7.8015 \cdot 10^{-4}$	$-2.5083 \cdot 10^{-5}$	$-1.5202 \cdot 10^{-7}$
0.25	2048	$7.5410 \cdot 10^{-5}$	$-2.2006 \cdot 10^{-7}$	$-1.2632 \cdot 10^{-10}$
1	128	$1.2324 \cdot 10^{-2}$	$6.9659 \cdot 10^{-6}$	$-1.4650 \cdot 10^{-6}$
1	2048	$1.2568 \cdot 10^{-3}$	$8.6240 \cdot 10^{-8}$	$-1.5362 \cdot 10^{-9}$
4	128	$1.0318 \cdot 10^{-1}$	$4.2705 \cdot 10^{-3}$	$8.6065 \cdot 10^{-5}$
4	2048	$1.1378 \cdot 10^{-2}$	$5.4963 \cdot 10^{-5}$	$1.2899 \cdot 10^{-7}$

Table 1: Executed coefficient diagnostics. These are finite numerical checks of the correction formulas, not proofs of their asymptotic order.

10.3 Cutoff profiles and a cancellation-free tail check

To compute the numerator in (2.9), restrict the sum in (10.1) to $j \leq M$ but leave p_0 unchanged. The ratio of the two target values is then precisely the finite coefficient fraction. Table 2

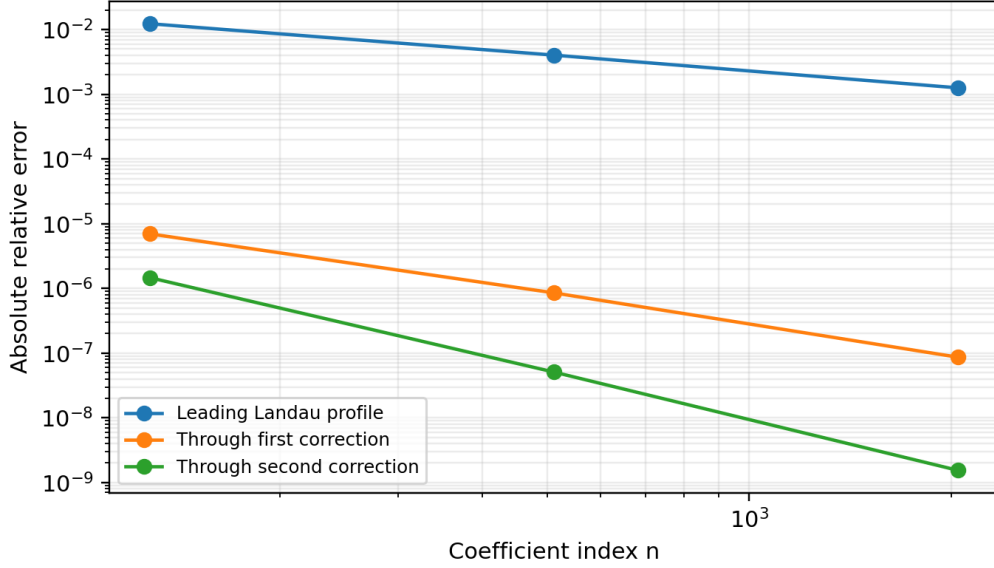


Figure 1: Absolute relative residuals for $\lambda = 1$, from the recorded positive recurrence and independent profile quadrature.

compares it with (8.9), evaluated at the actual rounded cutoff $m = M\delta$.

λ	M	$M\delta$	Finite fraction	Limiting fraction
1	175	0.5005791	0.4846012	0.4822650
1	350	1.0011582	0.9375042	0.9370268
1	699	1.9994559	0.9998768	0.9998716
4	55	0.5044705	0.0272113	0.0255110
4	109	0.9997688	0.5577570	0.5533841
4	218	1.9995376	0.9631579	0.9626222

Table 2: Cutoff diagnostics at $n = 2048$. Convergence need not be rapid for the smallest cutoffs; no equality of finite and limiting values is claimed.

The program also checks the first Palm moment (9.4) against the proposed equivalent in (9.6), using the rescaled endpoint variable from the proof. For $\lambda = 1$, the ratios of the numerically evaluated *first Palm moment* to that equivalent are

m	2	3	4	5
$J_1/\text{equivalent}$	0.8529573	0.9496008	0.9830888	0.9942918.

This does not directly numerically measure $1 - \mathcal{R}$ at extremely small loss. The passage from J_1 to that loss is proved by the factorial moment bound in Theorem 9.2. The numerical tail integral uses a finite upper endpoint in its rescaled variable; its truncation is not an interval-certified error estimate.

For the independent saddle check, $e^{\lambda\sqrt{2\pi\lambda}\mathcal{H}(\lambda)}$ is integrated in a Gaussian-scaled variable. The values of $\lambda(e^{\lambda\sqrt{2\pi\lambda}\mathcal{H}(\lambda)} - 1)$ at $\lambda = 10, 50, 100$ are approximately 0.0398933, 0.0412777, 0.0414697, consistent with the limit $1/24 \approx 0.0416667$. The verification record includes all inputs, not only these selected rows. Values of a cutoff profile printed as 1.0 can mean numerical saturation; for finite m the mathematical profile remains strictly less than one.

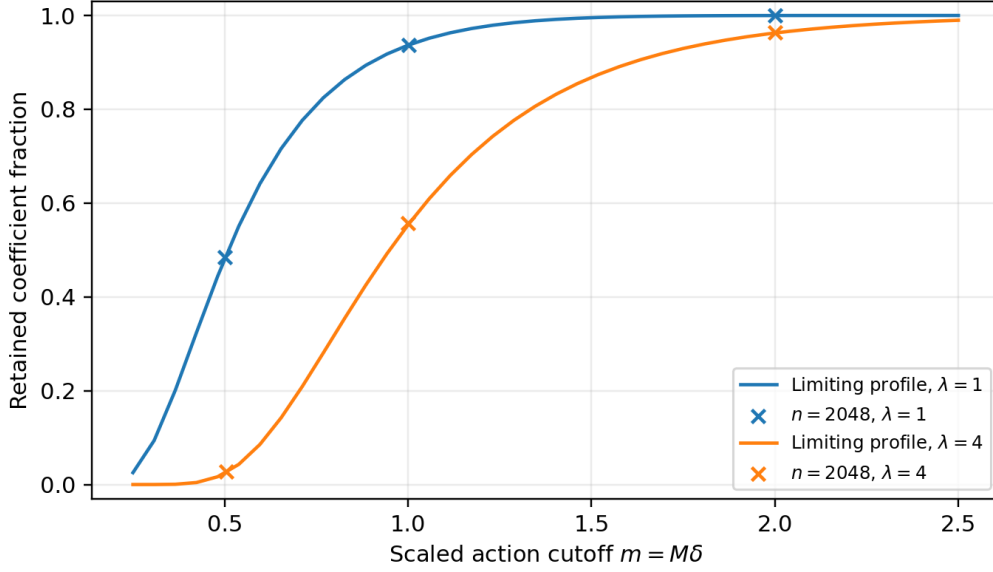


Figure 2: The limiting retention profiles and the independently computed finite coefficient fractions. All curves are floating-point evaluations of the formulas in this article.

11 Further research questions and concrete next steps

The following problems are not represented as solved by the results above. Each has a distinct mathematical target, rather than being a request to repeat the same calculation with renamed parameters.

11.1 The boundary-critical side of the lower endpoint

Our exact normalization assumes an interior fold. For $\varepsilon > 0$, the paths $c = 1/\zeta(1 + \varepsilon)$ and $c < 1/\zeta(1 + \varepsilon)$ lie outside it. Construct a uniform coefficient theorem that includes those paths and matches the present $\delta > 0$ chart. A natural starting point is the uncombined expression for $Q(x) - Q(0)$, retaining both the coupling mismatch and the nonanalytic power. The main difficulty is to find coordinates that remain informative when the exact positive saddle ceases to exist. This is the principal part of the repository’s lower-endpoint problem still left open here.

Editorial note (ProveIt, 2026-09-29). The path $c = 1/\zeta(1 + \varepsilon)$, and more generally $c = 1/(\zeta(1 + \varepsilon) + P'(1))$ for the tail $j^{-2-\varepsilon}$ changed by a fixed finite prefix P ($P = 0$ is the model of this article), is treated by two independent later packages (batch 52), `Lower_Critical_Endpoint_Cauchy_Cutoff_Condensation/lower_critical_endpoint.tex` and `Lower_Critical_Endpoint_Uniform_Coefficients_Compound_Poisson/article.tex`, for $\varepsilon_n \rightarrow 0$ at any rate. They prove the same theorem: a coefficient law uniform in every joint limit, which is the fixed- α critical law of [1] carried to $\alpha \downarrow 1$; one exceptional action whose deletion leaves the unconditioned configuration; for $n\varepsilon \rightarrow \infty$ a Landau cutoff profile; and for bounded $n\varepsilon$ a compound-Poisson deficit. The Landau law of the second package is the law with density p of (5.1); that of the first is its shift by $1 - \gamma$, γ Euler’s constant, and the two sets of cutoff quantiles coincide. At the fixed coupling $c = 1/\zeta(1 + \varepsilon)$ no interior fold exists, and neither package matches its results to the present $\delta > 0$ chart. The subcritical paths $c < 1/\zeta(1 + \varepsilon)$, a uniform theorem across $c = 1/\zeta(1 + \varepsilon)$, and that matching remain open. Paths are relative to `Analysis/Transseries/docs/series-and-transseries/`.

11.2 Growing cutoffs and finite-index tolerance certificates

Prove an error estimate for (8.4) that is uniform for $m = M\delta \rightarrow \infty$. To turn (9.8) into a useful finite-index certificate, the error must be small relative to $\exp\{-\lambda e^{m/\lambda}\}$, not merely $o(1)$. A promising route is to apply density-level insertion already in the finite Poisson model and compare the resulting negative-deviation local probabilities with their Landau counterparts. This avoids subtracting two nearly equal coefficient integrals. The relevant question is the largest joint range of m , δ , and the target loss for which the prefactor in (9.6) remains valid.

11.3 An all-orders conditional tail transseries

The proof of Theorem 9.2 can be continued by expanding both the Landau left tail and the endpoint substitution. Determine a closed recurrence for the coefficients multiplying $\exp\{-\lambda e^{m/\lambda}\}$, including powers of $e^{-m/\lambda}$ and $1/m$. The multi-jump terms live at much smaller scales involving $\exp\{-\lambda e^{km/\lambda}\}$. A complete hierarchy should state exactly how these scales interleave and should retain signs in the conversion from factorial moments to a probability.

11.4 Simultaneous small- λ matching

The small- λ formula (5.6) is a profile asymptotic. Develop a finite-coefficient theorem when $\lambda \rightarrow 0$ simultaneously with $\delta \rightarrow 0$, and identify its overlap with a single-large-action or condensation regime. The Fourier domination used in Theorem 4.3 loses strength as $\lambda \downarrow 0$, so a new global localization argument is needed. The right-tail logarithm $-1 - \log \lambda$ suggests the correct centering, but does not by itself prove the joint limit.

11.5 Finite-prefix and slowly varying tail perturbations

Replace $j^{-2-\varepsilon}$ by $j^{-2-\varepsilon}L(j)$, or add a finite polynomial to the polylogarithm. Determine which perturbations can be absorbed into the exact saddle, the amplitude A , and finitely many normal-form coefficients. For a finite prefix the singular part remains explicit; for a slowly varying L , the effective exponential scale can change. A useful target is a necessary-and-sufficient hypothesis under which the same Landau cutoff profile survives after normalization. No universality assertion for these modified models is made here.

Editorial note (ProveIt, 2026-09-29). At fixed $1 < \alpha < 2$ the slowly varying tail $aj^{-1-\alpha\ell}(j)$ is treated by a later sibling package, `Slowly_Varying_Action_Tails_Critical_Transseries/` (batch 50; its `thm:cutoff`, `thm:budget` and `cor:effective`), filed after this article was written. Its effective index $\alpha - m/\log b_n$ for $(\log j)^m$ tails is the interior analogue of the question posed here; the limit $\alpha \downarrow 1$ is not treated there.

11.6 The full hierarchy of Gaussian matching obstructions

Corollary 7.2 isolates the first threshold $\lambda\delta \asymp 1$. Subsequent thresholds involve the terms in $\lambda F_{0,\delta}(1)$, such as $\lambda\delta^2$. Develop a minimal renormalization theorem: for a specified joint range, which finite subset of fold-value terms must be retained to obtain a relative $1 + o(1)$ approximation? The exact convergent fold-value series provides the data; the new work is to combine it with an optimally uniform saddle remainder and to establish necessity as well as sufficiency of each retained scale.

11.7 Complex continuation and Stokes geometry

The inverse charts in Section 6 are local and their branches are fixed. Continue them around the logarithmic cut and identify the monodromy of the Landau contour representation. The fact that a Lambert function occurs in the leading inverse does not automatically identify all Stokes multipliers of the original countable-action problem. A satisfactory result should specify the complex parameter sectors, the admissible contour deformations, and which exponentially small differences correspond to distinct analytic solutions rather than to a change of normalization.

11.8 Arithmetic support and periodic action families

The full positive support, including action one, supplies the far-arc attenuation in Lemma 4.2. Determine the exact changes for supports of span greater than one, or for sparse regularly varying supports. In the periodic case a lattice factor and congruence restriction should replace the one-saddle formula. In genuinely sparse cases additional Fourier peaks may survive. An appropriate theorem should state an arithmetic condition on the support instead of silently reusing the aperiodic bound.

11.9 Validated computation at doubly exponential accuracy

Design a numerical evaluator for $\log(1 - \mathcal{R}(\lambda, m))$ with a rigorous absolute error bound, combining the positive Palm integral with a certified upper bound for the second factorial moment. The rescaled integral in the proof of Theorem 9.2 removes both underflow and severe concentration. Interval bounds for its tails would provide the missing validation layer. This is more useful than increasing the precision of a subtraction $1 - \mathcal{R}$ whose conditioning is already exponentially unfavorable.

11.10 A staged formalization rather than an unproved interface

An attainable first stage is a finite polynomial development of (5.12), the exact Lagrange identity in a formal power-series ring, and the first normal-form coefficients. A second stage can formalize the scalar analytic implicit chart and its two-sheet factorization. The uniform local-limit and Palm-tail theorems require substantial additional measure-theoretic and Fourier infrastructure. The repository's existing asymptotic-scale modules are relevant interfaces, but they are not a proof of these new analytical statements. A useful formalization should keep explicit the distinction between finite algebraic certificates and quantified asymptotic estimates.

12 Proof audit, provenance, and reproducibility

12.1 Logical dependencies and remaining boundaries

The main analytic input is the convergent polylogarithm continuation formula. The normal-form theorem is an exact algebraic cancellation applied to that formula. The coefficient theorem then requires three additional ingredients, all proved here: attenuation on all Fourier arcs, domination of the growing-contour remainder, and positivity of the limiting density. Its all-orders endpoint version uses a Taylor remainder under the same domination, not a purely formal interchange.

The inverse charts use the analytic implicit function theorem and Cauchy estimates. The Gaussian regime has a separate local-limit proof; its uniformity is not borrowed from the compact-window theorem. The cutoff theorem uses a killed characteristic function with the full centering retained. The sharp loss theorem uses a positive first Palm integral and a smaller

second factorial moment, so its prefactor is not inferred from an unconditioned tail or from a numerical fit.

All statements labeled theorems, propositions, lemmas, or corollaries in this article have conventional proofs. There is no claimed resolution of unrestricted transseries realization, general resurgence, or every path through the lower critical endpoint. No theorem in this package has been checked by Lean. Numerical assertions concern only the finite calculations explicitly recorded. An independent mathematical review is still needed before treating the draft as refereed research.

12.2 Repository snapshot and inspection scope

The inspected repository was VladimirReshetnikov/ProveIt, at commit

9250bbf8af80dacf7b252d9d0320b122c80961ba.

This is a commit identifier, not a claim that a mutable branch will always contain the same content. The connected GitHub tools were used to inspect the transseries directory tree, its project README, the series-and-transseries index, and selected package READMEs. The critical-Hahn package README was read in full. The beginning of its TeX article, including the exact parametrization, Lagrange coefficient, and conditional-Poisson formulas, was retrieved; the connector output for that long source was truncated. We do not claim a complete line-by-line audit of that article, the large canonical volume, or the whole repository. The recorded lower-endpoint question is explicit in the fully retrieved package README.

The recent inverse-digamma, Hahn–Fuchsian, finite-core, signed-quadratic, and non-Archimedean reversion package summaries were also inspected to avoid simply repeating an already filed direction. Their mathematical claims are not assumptions of any theorem here. The source file and README provenance is listed in the accompanying `notes/provenance.json`. No repository file, branch, issue, or pull request was modified.

12.3 Classical attribution and novelty status

The classical parts have been named explicitly: Lagrange inversion, the simply generated tree/conditioned-allocation connection, the polylogarithm expansion, the Landau law, the analytic implicit function theorem, saddle integration, and Poisson insertion. The proposed research contribution is their proved, explicit combination in the lower-endpoint model, particularly the confluent normalization, the matching obstruction, and the conditional cutoff equivalent. A targeted literature check was made, but it was not an exhaustive priority search. The present article does not certify that each formulation appears for the first time in the worldwide literature.

12.4 Rebuilding and rerunning

The package contains the LaTeX source `article.tex`, its compiled PDF, the figures and recorded data, the verification program, and a build script. The bibliography is embedded. From the package root, run

```
python -m pip install -r requirements.txt
python verify.py --output build/verification --max-n 2048
sh build.sh
```


The verification command does not overwrite the recorded `data/` outputs. The optional `make_figures.py` script regenerates the figures and its sampled profile data, and therefore does overwrite those files. Numerical outputs can differ in their final digits across platforms. On a platform whose `longdouble` has insufficient exponent range, the recurrence program raises an explicit error rather than silently returning an underflowed probability.

Editorial note (ProveIt, 2026-09-29). In the filed package `make_figures.py` writes by default under `build/` (`build/figures/` and `build/data/figure_curve.json`); only `-output-dir .` overwrites the shipped figures and `data/figure_curve.json`. The recorded `data/` files and `data/run.log` come from a run with `-output data`. The build report mentioned below describes the delivered build; this PDF was rebuilt on filing with editorial notes, so its recorded digests of the source and PDF no longer apply.

The theorem proofs do not require the Python dependencies. The PDF can be rebuilt from the stored figures with a standard mathematical LaTeX installation. The accompanying build report records the actual compilation checks, page count, rendered-page inspection, and file hashes for this delivered version.

13 Conclusion

The lower endpoint is singular in the coupling but regular after exact fold normalization. At $\alpha = 1$, the fold distance is $\delta \sim e^{-1/c}$, the sector crossover variable is $nc\delta$, and the normalized coefficients are governed by an explicit Landau profile. The same normalized phase has a convergent Lambert inverse chart and a uniform square-root chart at its moving critical value. The first failure of uncorrected Gaussian matching is quantified by $\exp(\lambda\delta/4)$.

Conditioning on the target coefficient has an even stronger effect on action truncation. The correct limiting budget is $M\delta$, and the loss at large scaled cutoff is

$$\frac{\sqrt{\lambda/(2\pi)}}{\mathcal{H}(\lambda)m^2} \exp\{-m/(2\lambda) - \lambda e^{m/\lambda}\}.$$

The prefactor and the double exponential follow from a positive Palm integral and a negligible multi-jump correction. These results provide a concrete interior-fold resolution at the repository's lower critical endpoint while making the remaining boundary-critical, finite-tolerance, and complex-continuation questions explicit.

References

- [1] VladimirReshetnikov/ProveIt, *Beyond Finite-Action Folds: Critical Hahn Transseries, Stable Sector Asymptotics, and Sharp Action Budgets*, package README and the beginning of `article.tex`, including the exact-coordinate propositions. Research package and editorial amendments, 29 September 2026. Pinned commit 9250bbf8af80dacf7b252d9d0320b122c80961ba. https://github.com/VladimirReshetnikov/ProveIt/tree/9250bbf8af80dacf7b252d9d0320b122c80961ba/Analysis/Transseries/docs/series-and-transseries/Critical_Hahn_Transseries_Beyond_Finite_Action_Folds.
- [2] VladimirReshetnikov/ProveIt, *Analysis/Transseries*, project README and documentation index, same pinned commit, inspected 29 September 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/9250bbf8af80dacf7b252d9d0320b122c80961ba/Analysis/Transseries>.
- [3] S. Janson, *Simply generated trees, conditioned Galton–Watson trees, random allocations and condensation*, Probability Surveys **9** (2012), 103–252; arXiv:1112.0510. <https://arxiv.org/abs/1112.0510>.
- [4] NIST Digital Library of Mathematical Functions, §25.12, *Polylogarithms*, especially Eq. 25.12.12 and its parameter restrictions. Inspected 29 September 2026. <https://dlmf.nist.gov/25.12>.
- [5] E. Bulyak and N. Shul’ga, *Landau distribution of ionization losses: history, importance, extensions*, arXiv:2209.06387v1 (2022). <https://arxiv.org/abs/2209.06387>. The density normalization used in the present paper is fixed explicitly by (5.1).
- [6] NIST Digital Library of Mathematical Functions, §4.13, *Lambert W-function*. <https://dlmf.nist.gov/4.13>.